

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 4th Year 1st Term Regular Examination, 2022
URP 4161
(Project Planning and Management)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Figures in the right margin indicate full marks.
ii) Answer any **three** questions from each section in separate script.
iii) Assume data where necessary.

Section – A

1. (a) Elaborate the concept- Vision, Mission, Policy, Strategy, Plan and Program with necessary examples. (13)
(b) Draw your concept on Project. Discuss types of project from multiple perspectives with necessary example. (12)
(c) What do you know about vision 2041 and SMART Bangladesh? (10)
2. (a) Draw your concept on project life cycle. (10)
(b) Describe the Initiation & Planning phases of project cycle management (13)
(c) Discuss various plans that are required to prepare in Project cycle management. (12)
3. (a) Distinguish between monitoring and evaluation with necessary example. (10)
(b) What do you mean by monitoring of a project? Discuss the objective of project monitoring and the types of monitoring with necessary example. (12)
(c) Draw your concept on project evaluation. Discuss various types of evaluation with necessary example. (13)
4. (a) Discuss various types of feasibility of a project with necessary example. (15)
(b) Discuss external evaluation and internal evaluation of a project? (10)
(c) Clarify the terms: Baseline, Endline, Audit, Review. (10)

Section – B

5. (a) What do you mean by Logical Framework Approach (LFA) based project design? Discuss the merits of an LFA based project design. (10)
(b) Who are the stakeholders of a project? Discuss the stakeholder analysis method with necessary example. (13)
(c) What are the strategies you might need to ensure stakeholder participation/engagement in a project? Discuss various approaches followed to minimize conflicts among stakeholders in a project. (12)
6. (a) A project's capital cost is estimated at \$ 5000 spread over three years. It is expected that \$ 3600 will be spent in the initial year and \$ 700 in each of the following two years. The benefit of the project is expected \$ 3000 for the 1st year and \$ 4000 for 2nd year. What are the Net Present Value (NPV) and Benefit Cost Ratio (BCR) of the project at 15% discount rate? What is the IRR of the project? (13)
(b) A company has secured a right to mine 100,000 tons of iron ore per (12)

year for 15 years. The current price per ton of iron ore is Tk. 3000 and it is expected to increase at the rate of 6% per year. What is the present value of the iron ore that you can mine if the discount rate is 16%?

- (c) Futura limited has an obligation to redeem Tk. 500 million worth of bonds 6 years hence. How much should the company deposit annually in a sinking-fund account where in it earns 14% interest to cumulate Tk. 500 millions in 6 year time? (10)
7. (a) Mention the basic principles of cash flow estimation. How is the time horizon for cash flow analysis usually established? (15)
- (b) Phoenix company borrows BDT. 500,000 at an internal rate of 14%. The loan is to be repaid in 5 equal instalments payable at the end of each of the next 5 years. Prepare the loan amortization schedule. (10)
- (c) Obtain the formula for estimating future value of an annuity; where A is an annuity, r is the compound interest rate, n is the duration of the project. (10)
8. (a) Define investment criteria. Mention the properties of Net Present Value in brief. (10)
- (b) A company is considering two projects, project M and project N, each of which requires an initial outlay of tk. 50 million. The expected cash inflows from these projects are: (25)

Year	Project M (Tk. In millions)	Project N (Tk. In millions)
1	15	27
2	12	20
3	25	15
4	30	13

If the cost of capital is 14%, calculate:

- Discounted payback period of M and N.
- IRR of M and N.

Also, discuss which project should the company invest in.

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 4th Year 1st Term Examination, 2022
URP 4141
(Environmental Planning and Management)

Full Marks: 210

Time: 3.0 hrs

- N.B.** i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Why environment is a system? Explain environmental system and various subsystems with necessary example. (13)
(b) What is meant by Urban Ecosystem Services? Discuss various urban ecosystem services with necessary example. (10)
(c) Discuss major global ecosystems/biomes with necessary example. (12)
2. (a) What do you mean by environmental management? Discuss the role of persuasive instruments in urban environmental management with necessary example. (10)
(b) What do you know about Command and Control Instrument (CCI)/Regulatory Instrument of environmental management? Discuss various CCIs with necessary example. (12)
(c) What is market-based instrument (MBI) of environmental management? Discuss various MBIs with necessary example. (13)
3. (a) Draw your concept on Professor Garrett Hardin's theory of Tragedy of Commons. Discuss the significance of tragedy of common theory in urban resource conservation. (10)
(b) What is meant by environmental externalities? Discuss various environmental externalities and internalization of environmental externalities with necessary example. (12)
(c) Distinguish between public goods and private goods. Illustrate various uses/values of non-market goods/services. Discuss the Contingent Valuation Method (CVM) of non-market environmental goods/services valuation. (13)
4. (a) What do you mean by linear flow of waste stream? Illustrate with necessary example. (10)
(b) How does environmental management make linear flow of wastes into a circular flow? Illustrate your response in light of the circular economy theory/perspective. (13)
(c) What do you understand by the term Sustainable Development? Discuss various dimensions and strategies of sustainable development. (12)

Section – B

5. (a) You are assigned to a project as an Environmental Planner and your task is to design mitigation measures for 'X' Chemical Industry. The industry emits toxic wastes including toxic chemical, heavy metals, and other harmful substances. Under some following condition, design a mitigation measure- process according to "Mitigation Measures Hierarchy". (15)
1. The industry is located in a remote area
 2. The industry is near settlements
 3. The industry is beside a river
 4. The soil type of the locality is 'less compact'
- (b) Briefly discuss the methods can be adopted for ensuring public participation in all stages of project. (12)
- (c) Write down some considerations you will keep in mind while choosing a method for impact identification. (08)
6. (a) Explain the attributes of impact with necessary example. (15)
- (b) How will you evaluate the project impact and degree of influence? Explain with proper example. (15)
- (c) Distinguish between site clearance and environmental clearance in case of EIA. (05)
7. a) What is Environmental Audit? Discuss various data and information needed for post EIA monitoring and auditing. (13)
- b) Discuss different types of stakeholders in an EIA system and their role with necessary example. (10)
- c) Prepare a hypothetical EIA report for an urban sector project in Khulna. (12)
8. a) What is the baseline situation? "Baseline data is the backbone of EIA" – justify the statement. (10)
- b) Correlate the steps of an EIA with project cycle. (13)
- c) What is environmental statement report? Why stakeholder participation is essential for an effective EIA operation? Discuss with necessary example. (12)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 4thYear 1st Term Examination, 2022
URP 4175
(Resettlement Planning)

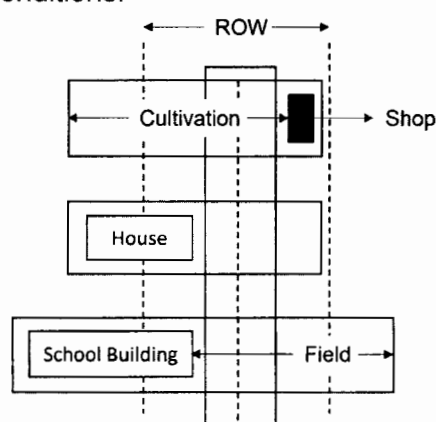
Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Write down the differences between displacement, resettlement, and rehabilitation. (09)
(b) If you are a "Resettlement Planner" then what safeguard issues will you follow while associated with an involuntary resettlement. (15)
(c) Write down the problems regarding to resettlement. (11)
2. (a) If resettlement is not done properly, what types of risk can happen. Discuss briefly. (10)
(b) Why Legal framework and Compensation given priority as first and second most important instruments of Resettlement Action Plan (RAP). (10)
Why do you think that they are given the right position on the priority list?
(c) Discuss about the facts regarding the compensation and design compensation for following resettlement conditions. (15)



3. (a) Correlate Resettlement plan with project cycle steps. (15)
(b) Write down the institutions involved in resettlement. (10)
(c) Why Resettlement Policy Framework (RPF) given more importance in Resettlement Action Plan? Write down the purposes of RPF. (10)
4. (a) Briefly discuss about the methods of valuing affected assets. (13)
(b) How you will redress the Grievances arise in the resettlement? (11)
(c) Mention the roles that can be played by project affected person in different stages of resettlement. (11)

Section – B

5. (a) "The present resettlement policies are adequate."- Give your arguments for or against this statement. (15)
(b) What are the powers of Deputy Commissioner and Arbitrator? (05)
(c) Provide recommendation to address the loopholes for finalizing land price in 1982 ARIPO Ordinance. (10)
(d) Differentiate between act, law, bill and ordinance. (05)
6. (a) Elaborate on the Risks to displaced persons/communities after resettlement. (15)
(b) Describe the various Resettlement Planning Instruments and Financial Assistance provided to the borrower. (15)
(c) What are the policy objectives of the World Bank on Involuntary Resettlement? (05)

7. (a) Why is economic analysis essential to Resettlement? (10)
- (b) Explain the limitations of Cost-Benefit Analysis for Involuntary Resettlement. (10)
- (c) Why is Property Right an issue in the Economics of Resettlement? (05)
- (d) Explain the No-Worse-Off criterion and its legality with respect to resettlement. (10)
8. (a) What are the advantages and limitations of Grievance Redress Mechanism? (10)
- (b) According to the IRR model, mention some of the risks and their counterparts. (10)
- (c) Describe the Pressure and Release (PAR) Model. (15)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 4th Year 1st Term Regular Examination, 2022
URP 4111
(Utility Services and Infrastructure Planning)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define Public Utility. Distinguish between Collective Utilities and Residential Utilities with examples. (10)
(b) Discuss the hierarchy of Collective Utilities with proper examples. (10)
(c) "Site and community specific characteristics must be considered for the provision of utility services" - Explain. (15)
2. (a) Illustrate the five components of Water Demand Management. (12)
(b) Discuss inelasticity of water demand with examples. In a residential area of 300,000 inhabitants, the specific water demand is estimated as 150 l/d, which includes leakage. Calculate the demand in 15 years' time if the assumed annual demand growth is 3.5%. Compare the results by applying the linear and exponential models. (13)
(c) Write a short note on Average Demand vs Specific Demand. Mention the factors affecting water demand of a geographical area. (10)
3. (a) Write a short note on "Needed Fire Flow (NFF)" with examples. (10)
(b) What are the basic rules of firefighting demand calculation and future water demand forecasting? (13)
(c) Estimate the fire flow rate and volume of water required to provide protection to a 10-story non-combustible building with an effective floor area of 8000 sq m. (12)
4. (a) Discuss the elements influencing sustainability of water supply in brief. (12)
(b) Illustrate various community managed water supply and sanitation systems in short. (12)
(c) Briefly discuss various water demand management measures with examples. (11)

Section – B

5. (a) Mention the demographic factors that can influence the infrastructure decision. (10)
(b) How do technology and fiscal constraints influence the decision of infrastructure investment? (10)
(c) Explain the role of the SDGs in determining and prioritizing investment decisions. (15)

6. Suppose, you have to choose a suitable site for the development of a major infrastructure. Based on Site Location, Available Labor, and Capital, there are four available alternative sites. The alternative sites are: A, B, C, D. Pair-wise comparison of the mentioned criteria and the ranking alternatives are given in the following tables: (35)

Criteria	Site Location	Available Labor	Capital
Site Location	1	$\frac{1}{2}$	3
Available Labor	2	1	4
Capital	$\frac{1}{3}$	$\frac{1}{4}$	1

Site Location				
Criteria	A	B	C	D
A	1	$\frac{1}{4}$	4	$\frac{1}{6}$
B	4	1	4	$\frac{1}{4}$
C	$\frac{1}{4}$	$\frac{1}{4}$	1	$\frac{1}{5}$
D	6	4	5	1

Reliability				
Criteria	A	B	C	D
A	1	2	5	1
B	$\frac{1}{2}$	1	3	2
C	$\frac{1}{5}$	$\frac{1}{3}$	1	$\frac{1}{4}$
D	1	$\frac{1}{2}$	4	1

Fuel economy	Cost in BDT (in million)
B	27
C	24
D	34
A	28

Select the best construction alternative using Analytical Hierarchy Process (AHP).

7. (a) Infrastructure decisions have some measures that go beyond benefit/cost analysis. Justify the statement for Padma Multipurpose Bridge Project. (15)
- (b) "The values of Social and Physical Infrastructures are linked with one another"- how so? Explain in brief with example. (10)
- (c) Why is it important to set up guidelines for infrastructure service delivery? Write down the differences between output and input based guidelines. (10)
8. Define the following terms (any five): (35)
- (a) Earmarked account
 - (b) Debt and equity
 - (c) Spillover benefits
 - (d) PPP
 - (e) Crowding in
 - (f) Crowding out
 - (g) Financing

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 4th Year 1st Term Regular Examination, 2022
URP 4171
(Legal Basis of Planning)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

- 1 (a) There is no significant difference among the terms “Act”, “Ordinance and “Rules”- Do you agree? Justify your answer. (05)
(b) Make a comparative analysis between major act and minor act. (10)
(c) Briefly discuss the problems of minor act. (10)
(d) Elaborate the importance of legal aspects of planning in a developing country like Bangladesh. (10)
- 2 (a) “Land acquisition is the result of the application of the eminent domain power of a state” – Explain. (05)
(b) Make a comparison between the procedures of normal acquisition and emergency acquisition. (15)
(c) Explain the regulatory intervention in land acquisition process in our country with necessary example. (15)
- 3 (a) State the factors on the basis of which compensation for the acquisition of land is determined. (15)
(b) “Involvement of different authorities is a major drawback in land acquisition” – Explain with necessary examples. (10)
(c) Write about the penalties imposed on the violation of regulations associated with land acquisition. Explain the importance of its necessary re-arrangements. (10)
- 4 (a) Briefly discuss the requirements for getting the approval of a private housing project according to “Private Housing Project Land Development Rule’2004”. (10)
(b) Consider yourself an urban planner employed by a real estate firm. A layout plan has to be created for a private housing project. Describe in detail what must be in the layout plan for the project to be approved by the concerned authority. (12)
(c) Specify the direction of “Private Housing Project Land Development Rule’2004” regarding land ownership, rehabilitation of affected people, and project area saleable land. (08)
(d) Make a list of clearance certificates necessary to submit for the approval of private housing project. (05)

Section – B

- 5 (a) Discuss the important planning aspects that have been duly considered in the BNBC 2020. (20)
(b) Formulation of BNBC is a huge task of multi-professional experts. Evaluate this statement with appropriate examples. (15)

- 6 (a) As an Urban Planner critically analyse the consequences and remedies of building deviations in the urban areas of Bangladesh. (20)
- (b) Discuss the governance and functions of RAJUK. (15)
- 7 (a) Discuss with examples the importance of coordinated efforts of many interrelated ministries and departments for the formulation and implementation of URP Act 2017. (15)
- (b) Narrate the functions of the Urban and Regional Planning Executive Council under the draft URP Act 2017. (10)
- (c) Discuss the duties and responsibilities of Planning Inspectors in terms of URP Act for any country. (10)
- 8 Write notes on the followings (any five) (35)
- (a) Establishment of the Municipality;
 - (b) Ward Committee;
 - (c) Sources of fund for a municipality;
 - (d) Master Plan of a municipality;
 - (e) Site development schemes;
 - (f) Regulation of buildings;
 - (g) Development plan.